

Trainings ▾

Preparatory Steps

⚠ CAUTION ⚠

As far as possible the pump should be dismantled while the engine timing pin is engaged and piston number 1 is at the TDC position.

⚠ CAUTION ⚠

If space does not permit to do the same, pump can be taken out in any position of engine crank.

⚠ CAUTION ⚠

Pump should be fitted in locked condition (pump timing pin engaged in cam lock mechanism on governor rotor) and engine piston No.1 position at TDC. (Engine timing pin engaged).

⚠ CAUTION ⚠

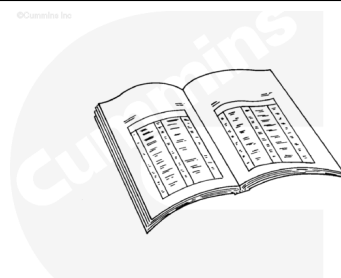
Immediately after dismantling pump, bar the engine to achieve piston No.1 position at TDC and engine timing pin engaged. Do not forget to disengage the timing pin. Trying to bar the engine while timing pin engaged will break away timing pin tip.

⚠ CAUTION ⚠

Protect the fuel pump openings immediately by protective caps available with Service Engineer.

⚠ CAUTION ⚠

Cleanliness during the operation is most important. Dirt particles will damage feed pump suction valve and dirt in fuel gallery will damage pump elements seriously.



ck800wa

⚠ CAUTION ⚠

Do not use cotton waste.

⚠ CAUTION ⚠

Always perform operation with proper tools and spanner. Do not use inch spanners at all.

⚠ CAUTION ⚠

In absense of gear pullers do not hammer pump shaft while taking out pump assembly from place. This will seriously damage threading on pump camshaft.

⚠ CAUTION ⚠

If you are replacing a pump store all protective caps and caution tags with you.

⚠ CAUTION ⚠

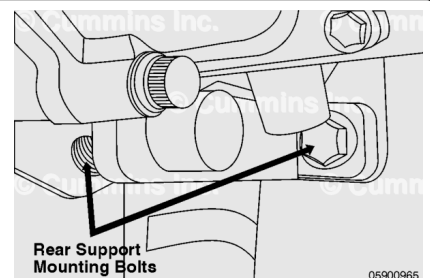
Do not lift the pump by holding feed pump. Support it from below governor housing and front flange.

Remove

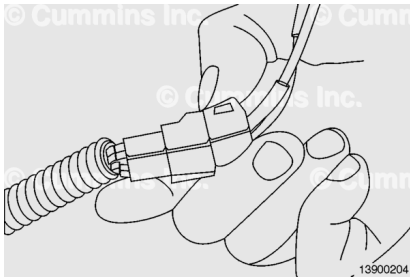
Dismounting Motorpal fuel injection pump.

Remove rear support mounting bolts and bracket to fuel pump assembly.

Size : 14mm



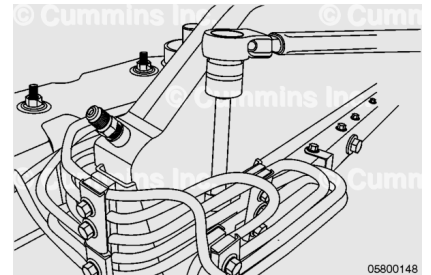
Unplug shutoff coil lug from wiring harness.



Removal of L.P. piping.

Remove all fasteners for various clamps.

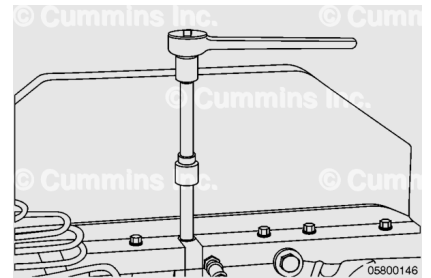
Size : 10mm



Removal of L.P. piping.

Remove fuel supply and drain connections on feed/fuel pump

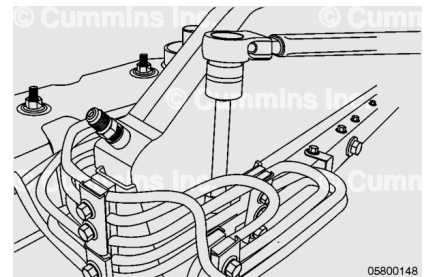
Size : 19mm



Removal of H.P. piping

Remove H.P. pipe clamping bolts on intake manifold/aftercooler.

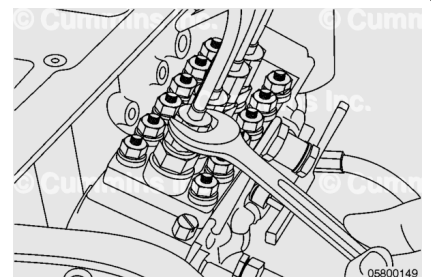
Size : 10mm



Removal of H.P. piping.

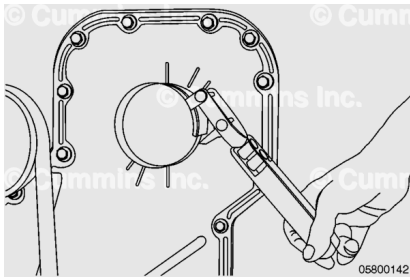
Loose all end connections fitted to fuel pump delivery nipples

Size : 19mm and 17mm for 140 KVA



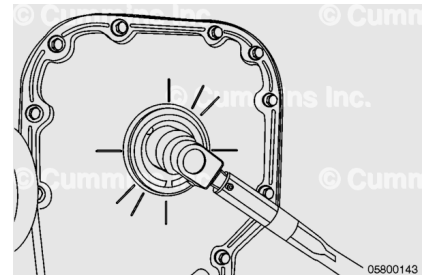
Remove access cover to drive gear.

Filetr Wrench Size : 90-95mm



Loosen the drive gear retaining nut.

Size : 27mm

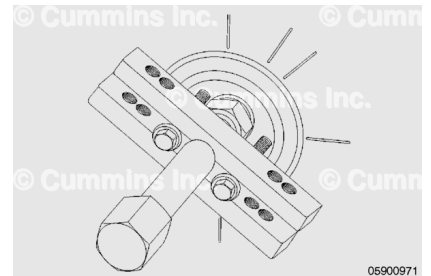


Pull out the drive gear.

Gear Puller Part Number : ST-647

Socket : 25mm

Bolts : M8 x 1.25 x 80mm



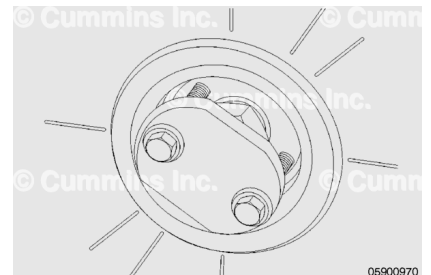
Alternative method.

Pull out drive gear.

Gear Puller Part Number : 3820069

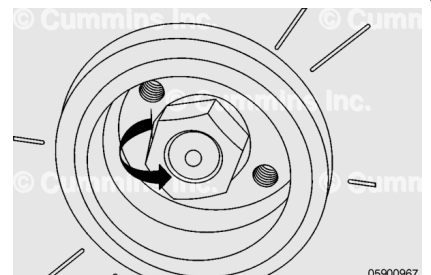
Socket : 10mm

Bolts : M8 x 1.25 x 50mm



⚠ CAUTION ⚠

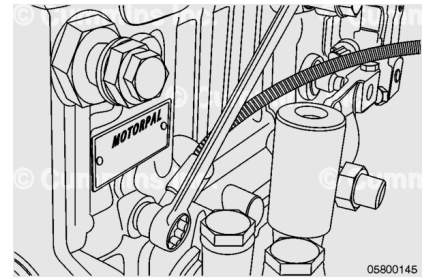
Take out drive gear nut and lock washer very carefully not to fall down in oil sump.



Dismount the lube oil supply hose to the fuel pump.

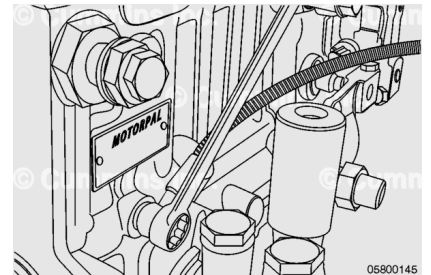
Retain banjo M8 x 1.25 in place.

Open end spanner size : 12mm



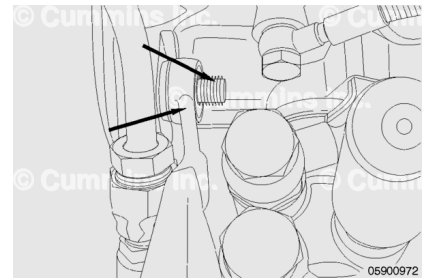
Loosen nut for hose oil supply at cyl. block end if required.

Size : 14-15mm



Loosen fuel pump outboard nuts and store

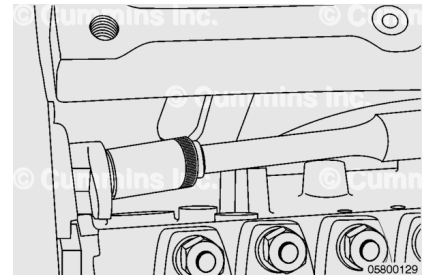
Ring Spanner Size : 14-15mm



Loosen inboard pump mounting nuts and store.

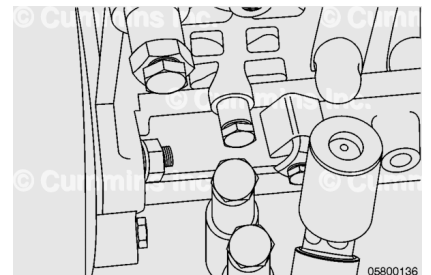
Service Tool Number : 3823956

Size : 14-15mm



Pull out pump assly. and let rest on studs. Carefully pull out completely and store on clean surface.

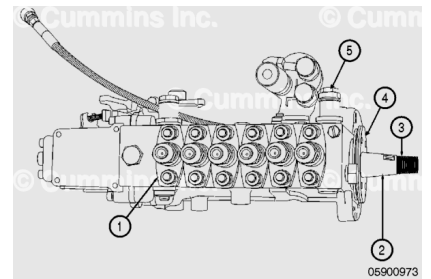
Immediately put on protective covers on all opening ports to prevent dust entry.



Inspect for following

- Protective Cover
- Camshaft Nose Surface

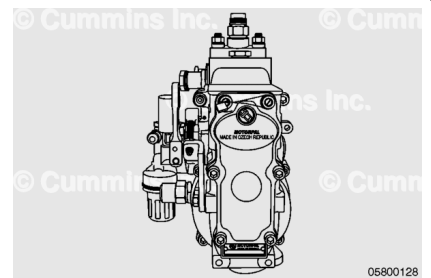
- Drive Gear Nut and Threads
- O-Ring Pump Mounting
- All Adaptors/Banjos



Install

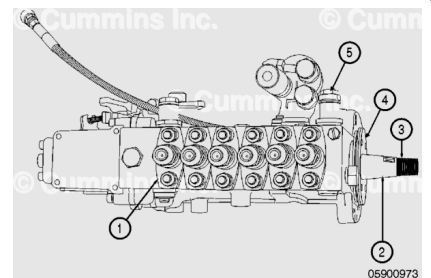
Mounting Motorpal fuel pump on engine.

Ensure that pump is in timing lock condition. If not, follow the pump timing lock procedure.



Inspect for following

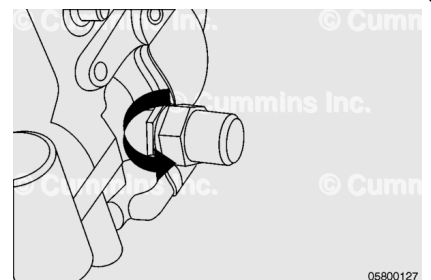
- Protective Cover
- Camshaft Nose Surface
- Drive Gear Nut and Threads
- O-Ring Pump Mounting
- All Adaptors/Banjos



Procedure to lock fuel pump for timing.

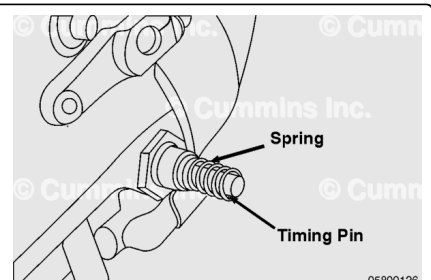
Remove cover timing pin

Size : 22mm



Position of timing pin and retaining spring should be as shown.

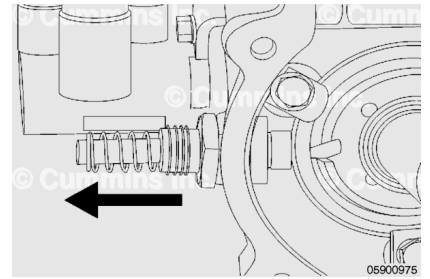
This is the position for the timing pin disengaged; springs in front of the timing pin.



Spring in front of timing pin timing pin disengaged

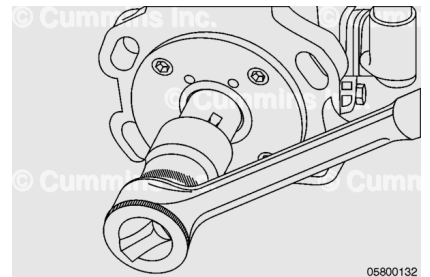
Actual condition inside governor.

Take out timing pin and spring.



Assemble nut and lockwasher on pump camshaft. Rotate camshaft clockwise looking from drive end.

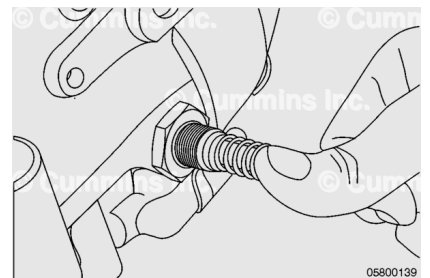
Size : 27m



Spring behind timing pin. Timing pin engaged.

While rotating camshaft, insert timing pin. Put spring behind timing pin and lightly press by thumb.

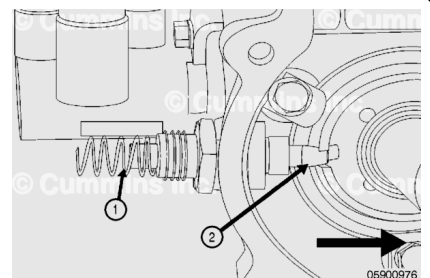
At one position of camshaft, timing pin slides into notch provided on governor rotar assembly.



Spring behind timing pin.

Timing pin engaged.

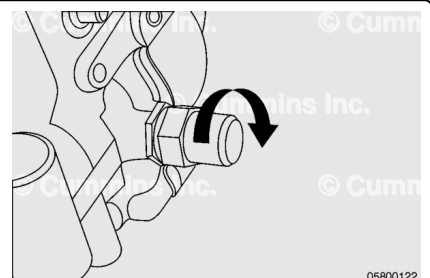
Do **not** rotate pump after timing pin is engaged.



Install timing pin cover.

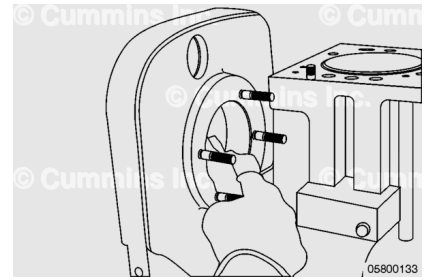
Torque Value: 20 n•m [15 ft-lb]

Size : 22mm



Mounting Motorpal fuel pump on engine.

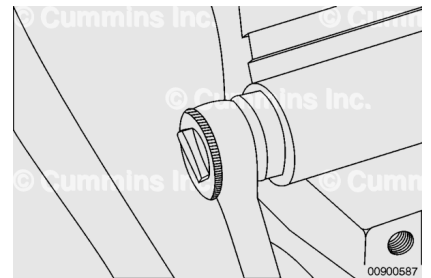
Clean mounting surface of fuel pump in gear end housing by soft clean cloth.



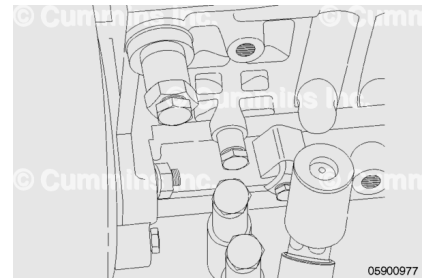
Bar the engine with the help of barring tool. Simultaneously press the timing pin against the camshaft gear.

Bar Tool : 3377371

Ratchet : ½ inch square drive



Locate the nuts for fuel pump and hand tight. Slide drive gear completely on the fuel pump camshaft.

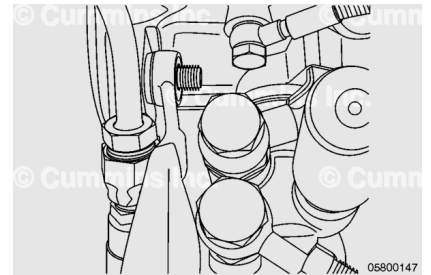


Outboard mounting nuts.

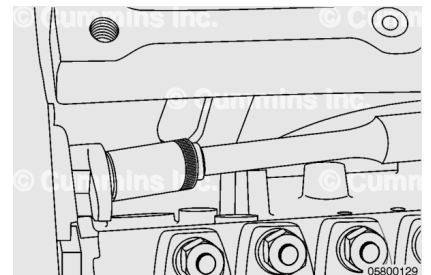
Tighten inboard and outboard mounting nuts.

Torque Value: 30 n•m [22 ft-lb]

Size : 14-15mm



Inboard mounting nuts.

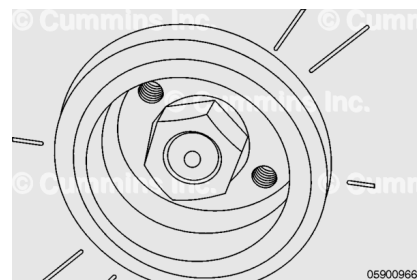


Do not overtighten.

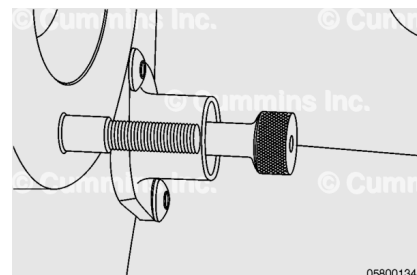
Install and tighten pump retaining nut and lock washer.

Torque Value: 15 n•m [11 ft-lb]

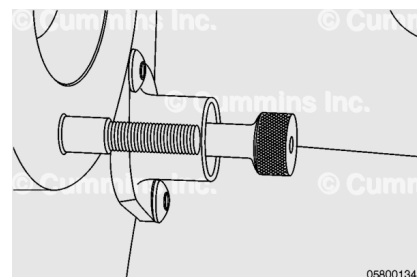
Size : 27mm



Disengage the engine timing pin. Pull out.



Engine timing pin engaged.

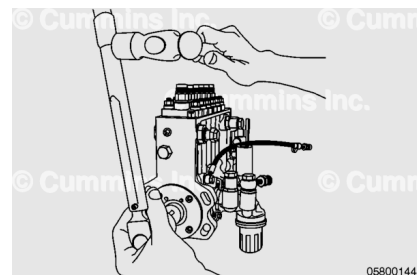


Remove drive nut on pump camshaft (pump timing pin in engaged position). Strike a wrench with sharp blow in anti-clockwise direction.

Socket : 27mm

Ratchet

Plastic Hammer

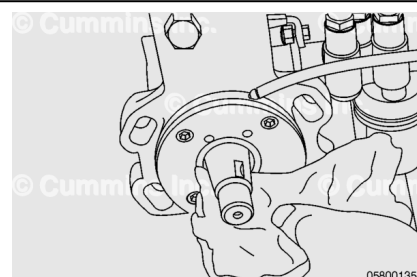


Clean surface of camshaft by soft cloth.

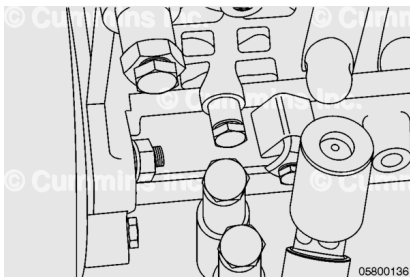
Lubricate o-ring on pump mounting front cover. Replace o-ring if damaged.

Oil Can/Dispenser

Clean Soft Cloth

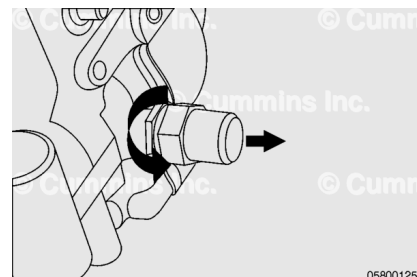


Locate the fuel pump in the pilot bore of gear end housing.



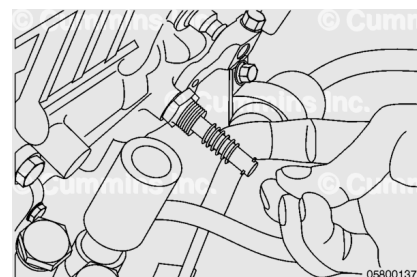
Remove fuel pump timing pin cover.

Size : 22mm



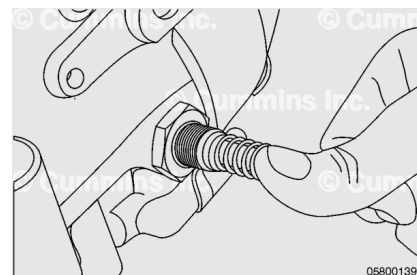
Spring front of timing pin

Remove timing pin along with the spring. (Spring in front of timing pin)



Spring behind timing pin

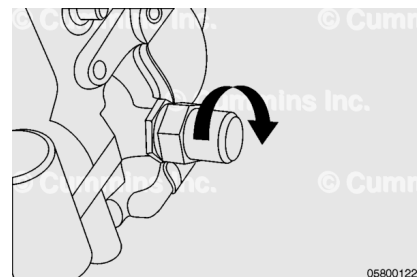
Change position of spring from front to behind the timing pin.
This will unlock the pump.



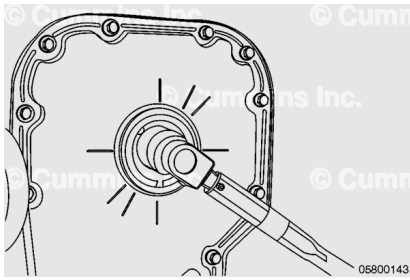
Install timing pin cover and torque.

Torque Value: 20 n•m [15 ft-lb]

Size : 22mm



Torque fuel pump drive gear nut in.

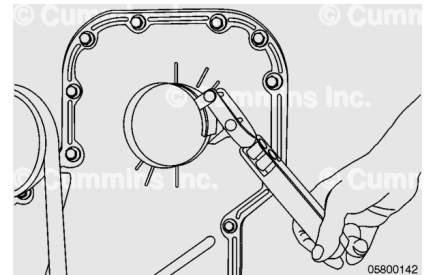


⚠ CAUTION ⚠

Do not overtighten

Install gear cover access cap.

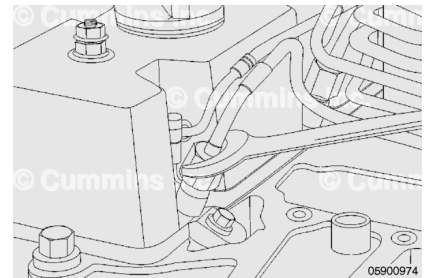
Wrench Size : 90-95mm



Tighten H.P. pipe end nuts on delivery nipples and torque.

Size : 19mm

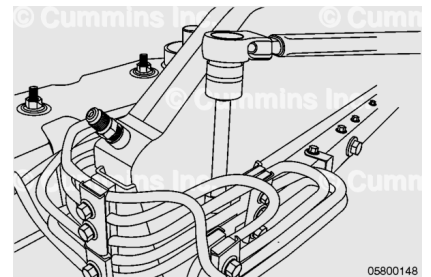
Torque Value: 24 n•m [18 ft-lb]



Tighten H.P. pipe holding clamp and torque.

Torque Value: 24 n•m [18 ft-lb]

Size : 10mm

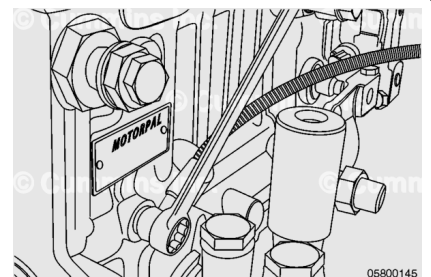


Assemble lube oil supply pipe to fuel pump and tighten.

Replace sealing washer.

Torque Value: 10 n•m [7 ft-lb]

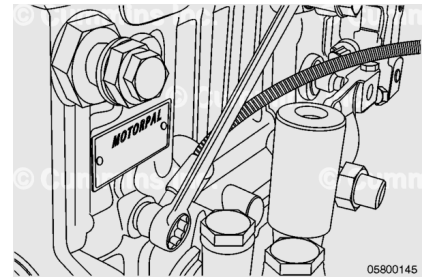
Size : 12mm



Assemble lube oil supply pipe to elbow on cyl. block.

Hand tight moderately.

Size : 14mm



Assemble all low pressure pipes. Replace all sealing washers for banjos on fuel pump.

Size : 19mm

Torque Value:

Banjo on fuel pump 24 n•m [18 ft-lb]

Torque Value:

Banjo on engine 30 n•m [22 ft-lb]

Remove oil fill plug on fuel pump body.

Fill up 360ML of fresh engine oil (API 15W-40).

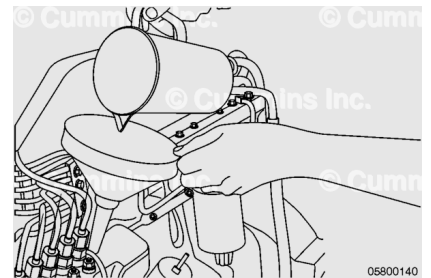
Install plug.

Clean container 1 Liter

Clean Plastic Funnel

Size : 19mm

Torque Value: 30 n•m [ft-lb]



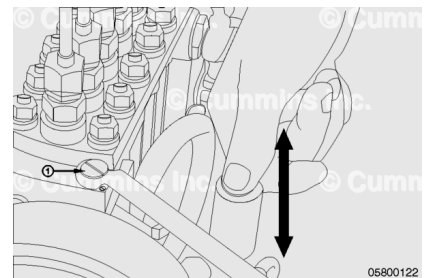
Loose bleed screw. Prime fuel by hand prime pump with strokes as shown.

Bleed all air from fuel system up to fuel pump.

Tighten bleed screw.

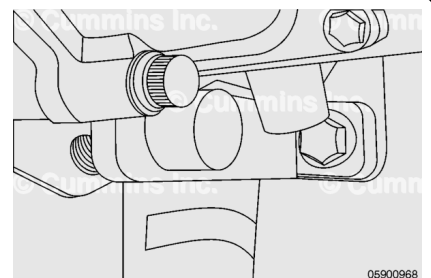
Size : 10mm

Torque Value: 9 n•m [7 ft-lb]

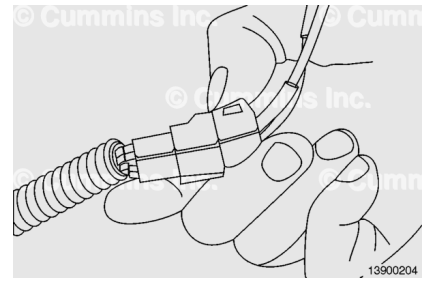


Assemble rear fuel pump support

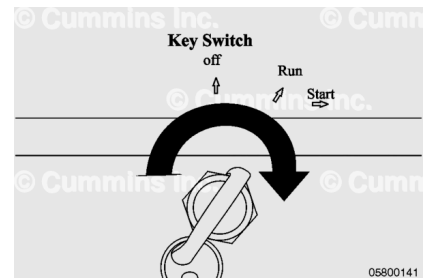
Size : 14mm



Plug in lug for shut off coil



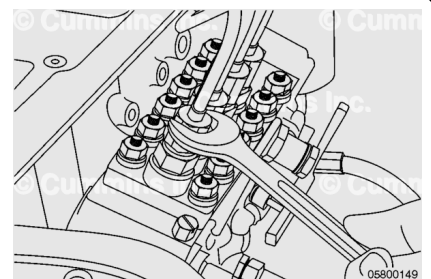
Turn ignition key switch on and hold up to crank position.
Crank the engine.



Shutoff coil will be in energized condition.
Shutoff lever is appx. horizontal.

©Cummins Inc.

Slacken the end nut for H.P. pipe for injector No.1
Crank the engine and bleed air from H.P. pipe till clear fuel
squeezing out of it. Tighten the H.P. pipe end nut.
Start the engine at low idle.
Size : 19mm



Torque Value: 24 n•m [18 ft-lb]